

TIN DANG HUYNH TRUNG

Senior Software Engineer – Backend, AI Platforms & Distributed Systems

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Senior software engineer with eight years of building and running production backend systems in Python, Rust and TypeScript across fintech, conversational AI, computer vision and data platforms. Sole architect on several greenfield platforms, technical lead on teams of up to fourteen. I lean toward systems where correctness is enforced by structure rather than convention: append-only ledgers, multi-tenancy that fails closed, money that cannot move twice.

PROFESSIONAL EXPERIENCE

Senior Software Engineer / Technical Lead

Trustify Technology

📅 Jan 2024 – Present

📍 Ho Chi Minh City, Vietnam

I architect and lead client rebuilds across Vietnam and Myanmar — fintech, real estate and intelligent transport for Yoma Group, Yoma Bank, Wave Money and Trustonic — and wrote the company's engineering career ladder.

- **Estate-finance platform** • *Yoma Group — sole engineer & architect*

The legacy system kept balances in rows anyone could edit, so I rebuilt it off PHP/Laravel onto a single append-only double-entry ledger where every balance can be re-derived and audited, with per-customer cutover, an outbox and reconciliation around it. Before anyone moved I graded 235 legacy business rules against the new system and published the gaps.

Python 3.12 • FastAPI • async SQLAlchemy 2 • PostgreSQL (schema-per-tenant) • Redis • Next.js • AWS ECS • Envoy

- **Card payments & PCI boundary** • *Yoma Group, Wave Money*

I put cardholder data behind its own service, database and network so that only that service falls inside PCI scope, and made the build reject any import that crosses the boundary. Card payments run through 2C2P with callbacks that are safe to retry and a three-way reconciliation behind them.

Python • FastAPI • PostgreSQL • 2C2P • Docker

- **Urban traffic intelligence** • *Ho Chi Minh City — architect & sole engineer*

400 cameras produce more video than a city network can carry, so I kept the footage at the edge and sent only events onward. Jetson nodes cut backhaul by 75%, YOLO models on TensorRT track vehicles at a P95 of 85 ms, and events travel to the data centre over mutually authenticated Kafka. I load-tested the pipeline to 71K events a second before adopting it.

Python • FastAPI • Kafka • PyFlink • TensorRT • Triton • K3s • ArgoCD • Jetson • DuckDB

- **Microfinance lending & resident services** • *Yoma Bank, Star City*

I built Yoma Bank's microfinance lending MVP and led the bilingual staff-culture survey alongside it. On Star City's resident platform I owned the money, authentication and payment paths for a team of nine, and closed auth-enumeration, TOCTOU and privilege-escalation findings before launch.

Java 21 • Spring Boot 3.3 • jOOQ • Kafka • Flyway • Python • FastAPI • TypeScript

- **Device-management migration** • *Trustonic*

A customer could only be moved if they could be moved back, so I owned the rollback across PostgreSQL and DocumentDB. I tuned the migration jobs as well: vectorised batch validation, broadcast joins in place of shuffles, keyed DynamoDB reads so a run never scans a live table, and Redis partitioning chosen by measurement.

PySpark • AWS Glue • Airflow • DynamoDB • PostgreSQL • DocumentDB • Redis • boto3

Founder & Principal Engineer

Pilot Space (open-source AI infrastructure)

📅 Jan 2026 – Present

📍 Remote

- **Moon** • *Redis-compatible in-memory store, written from scratch*

I wrote it from scratch so that existing Redis clients connect unchanged: RESP2/RESP3, 250+ commands, and native vector, BM25 and graph search under MVCC. It has shipped 30 releases to v0.8.9 and measures 1.62× on GET and 1.32× on SET against Redis 7.0 with 8–22% less memory per key, published alongside the cases where it loses.

Rust • Tokio • io_uring • crates.io • PyPI

- **Lunaris & Helios** • *agent memory and agent harness*

Lunaris recalls across 100K documents per scope in 19–22 ms at the median, and a CI gate fails the build on any 5% regression. Helios is the harness that consumes it: 12 provider backends, a native MCP client and server, WASM-sandboxed extensions, and a circuit breaker on every outbound call.

Rust • Tokio • MCP • Wasmtime • npm

- **Hydroa** • *multi-tenant LLM gateway*

A gateway that meters what each tenant spends on LLMs. Envoy authorises at the edge, six providers sit behind per-tenant and per-provider circuit breakers, customer keys are encrypted at rest, and a cost ledger reconciles against what the providers actually billed.

Python • FastAPI • Envoy • PostgreSQL • Kubernetes • OpenTelemetry

- **ADD** • *published development method*

Specify the work, write the failing tests, build until they pass, then verify against a recorded result rather than a plausible diff. Published as @pilotspace/add and pilotspace-add at about 1.2K downloads a month.

CTO – Co-founder

ClickAI Viet Nam

📅 Jun 2024 – Oct 2025 (concurrent)

📍 Ho Chi Minh City, Vietnam

I co-founded ClickAI (clickai.vn) and built its conversational AI platform from scratch — multi-modal chat, image and audio generation, community — to 40K+ users across Southeast Asia inside four months. As CTO I set the architecture and engineering standards, led a team of twelve to fourteen, and owned billing, identity and deployment myself.

- **Billing & metering** • *sole author*

I built the wallet, plan, promotion and referral-commission engine over a month-partitioned ledger that meters every LLM call. A reconciliation check I wrote found provider billing had drifted more than 30% from what we metered; I corrected the ledger model and its indexes, keyed idempotency on the source event so a retry cannot double-count, and made the check a standing job.

Python · FastAPI · PostgreSQL (partitioned) · Redis

- **Model platform** • *20+ providers*

I built the orchestration layer that routes across 20 third-party providers and one of our own, with failover, vector memory and reasoning-model support; choosing the model by policy rather than habit cut API spend by 20%. Generation fans out to ComfyUI, Replicate, fal.ai and ElevenLabs behind circuit breakers.

Python · FastAPI · Weaviate · Qdrant · FalkorDB

- **Production, identity & web** • *99.9% uptime, 1K+ concurrent sessions*

I authored the 13-service deployment with cert-manager and autoscaling, and held sub-2s responses. When the database saturated I moved PostgreSQL onto a three-node Patroni cluster; when a release regressed, my own instrumentation caught it before any customer did; and a hosting comparison I took to stakeholders halved the monthly bill. I also shipped OAuth, TOTP and refresh-token rotation, and the web client's community and creative features.

Kubernetes · Helm · Azure · Patroni · Prometheus · Next.js 14 · TypeScript

Earlier Experience (2018 – 2024)

- **Senior Data Engineer**, Asia Commercial Bank (ACB) • Aug 2023 – Jun 2024

Ran an enterprise data lake carrying 5M+ daily transactions. A data-quality framework I built across six dimensions cut manual verification by 80%, and my APIs served 100+ internal applications. Cloudera · PySpark · Dagster · Iceberg · FastAPI

- **Software Data Engineer**, Vietmoney Corp • Mar 2021 – Aug 2023

Built on a fintech appraisal and lending platform running 10M+ daily transactions at 99.9% uptime. Price forecasting at 96% accuracy cut appraisal from 15 minutes to 5 seconds, Vietnamese NER reached 96% F1, and a distributed crawler doubled throughput across 50+ sources.

Python · PyTorch · BERT · Scrapy · Pub/Sub · BigQuery · Airflow · Kubernetes · GCP

- **Software Developer**, Dai Phat Solutions • Aug 2019 – Mar 2021

Shipped CCTV analytics at 15 FPS to 20+ clients, cutting reported incidents by 40%, and face recognition at 92% accuracy over 1,000+ employees that removed 90% of a manual process. Python · PyTorch · OpenCV · FAISS · Kafka · FFmpeg

- **Full-stack Developer**, LogiGear Corporation • Jun 2018 – Mar 2019

Built wallet, smart-contract and liquidity-pool management on Ethereum with MetaMask, in Node.js and React.

OPEN SOURCE

- **Moon**, **Lunaris** and **ADD** at github.com/pilotspace, described above. **yowo** (PyPI) — production YOLO inference and export across TensorRT, ONNX Runtime, CoreML and OpenVINO, with ByteTrack and CLIP-ReID tracking. **fastapi-multi-tenant**; **rjson**, Rust-backed JSON for Python; **face-anti-spoofing**, founded at Vietmoney.

TECHNICAL SKILLS

Languages & backend: Python (typed, async), Rust, TypeScript / Node.js, SQL, Java; FastAPI, SQLAlchemy 2, Next.js / React, clean & hexagonal architecture, CQRS, event sourcing, multi-tenancy, outbox & idempotency, PCI boundaries, OAuth / OIDC

AI, data & cloud: PyTorch, YOLO, TensorRT, Triton, LLM gateways, RAG, MCP; Kafka, PyFlink, PySpark, AWS Glue, Airflow, BigQuery; PostgreSQL (Patroni HA, RLS, partitioning), Redis, MongoDB, DynamoDB, Weaviate, Qdrant; Kubernetes, Helm, Azure, AWS, GCP

EDUCATION, TEACHING & CERTIFICATIONS

BSc Computer Science (High-Quality Program), Ton Duc Thang University, 2015–2019, GPA 3.2. **Teaching:** Design Patterns instructor, Ton Duc Thang University (2022); mentor, MindX (2023); speaker, AI Workshop, FPT University (2025). **Certifications:** Google Cloud AI Agents (2025); AWS ML Foundations (2022); Scrum (2017). **Languages:** English, Vietnamese.